3-BIT FLASH ADC USING TIQ COMPARATOR

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***Abstract*—** This paper discusses the design and implementation of a 3-bit Flash Analog-to-Digital Converter (ADC) using Threshold Inverter Quantization (TIQ) comparators in 90nm CMOS technology. TIQ comparators provide a small and low-power option compared to traditional comparators by taking advantage of the variable threshold behavior of CMOS inverters. We designed seven TIQ comparators with carefully adjusted switching thresholds to divide the input analog voltage range into eight levels. These comparators produce a thermometer code, which a logic gate-based priority encoder then converts into a 3-bit binary output. We implemented and simulated the complete system in Cadence Virtuoso using 90nm technology. The resulting design shows high-speed operation with a delay of 40 µs and low power consumption of 300 µW. This work demonstrates the potential of TIQ-based Flash ADCs for compact and energy-efficient signal processing applications.

Keywords*— *Flash ADC, TIQ Comparator, CMOS 90nm Technology, Threshold Inverter Quantization, Priority Encoder, Low Power Design.

I. INTRODUCTION

Analog-to-Digital Converters (ADCs) are essential in modern electronic systems. They convert real-world analog signals into digital format for processing, storage, or transmission. Among various ADC designs, the Flash ADC stands out for its speed, making it ideal for high-frequency applications like digital oscilloscopes, radar systems, medical imaging, and wireless communications. However, the traditional Flash ADC design has significant drawbacks. It consumes a lot of power, takes up a large chip area, and has increased input capacitance, especially with higher resolution. A 3-bit Flash ADC requires 7 comparators, along with a resistor ladder to create reference voltages and a complex encoder to process outputs. This design becomes inefficient in terms of area and power as the resolution increases. To tackle these issues, a modified design based on Threshold Inverter Quantization (TIQ) has been proposed and implemented in this work. The main idea of TIQ is to adjust the threshold voltage of a CMOS inverter by changing the width-to-length (W/L) ratio of the PMOS and NMOS transistors. By sizing these transistors carefully, each inverter switches at a specific voltage threshold. This approach eliminates the need for resistor ladders and complex comparators. In this project, a 3-bit Flash ADC is designed using seven TIQ comparators. The switching thresholds range from 0.2V to 0.8V in steps of 0.1V. These thresholds are set by modifying the W/L ratios of the PMOS and NMOS transistors in each comparator, allowing the input voltage range to be divided into eight distinct levels. Each comparator outputs a logic high ('1') when the input exceeds its threshold voltage and logic low ('0') otherwise, generating a thermometer code. The outputs of these TIQ comparators feed into a 3-bit priority encoder, which is built using basic logic gates such as AND, OR, and NOT gates. The priority encoder's job is to identify the highest-order active comparator and produce the corresponding binary output, effectively turning the thermometer code into standard binary code. The encoder's logic was derived using truth tables to ensure minimal delay and accurate digital representation. The entire ADC was implemented and simulated using 90nm CMOS technology, and testing confirmed correct functionality across all input voltage levels. This work details the step-by-step methodology of TIQ comparator design, setting threshold voltages through transistor sizing, designing encoder logic, and presenting simulation results. This makes the TIQ-based Flash ADC a practical and efficient alternative to traditional designs, especially where power, area, and speed matter. In conclusion, this project showcases the design and implementation of a 3-bit Flash ADC using TIQ comparators and a logic gate-based priority encoder, achieving a balanced trade-off between speed, power, and silicon area. This report further discusses transistor-level sizing, delay analysis, and power measurement to validate the proposed ADC architecture as a viable solution for modern integrated systems.To design a 3-bit Flash ADC using a TIQ comparator in 90nm CMOS technology with an emphasis on minimizing power consumption and propagation delay. This work aims to design and implement a 3-bit Flash ADC using TIQ comparators and a logic gate-based priority encoder optimized for low power, high speed, and small chip area. By varying the W/L ratios of the transistors, the switching thresholds of seven TIQ comparators are set between 0.2V and 0.8V in 0.1V steps. The thermometer code output from the comparators is converted to binary using a custom-designed encoder made with basic logic gates. The entire design is developed using 90nm CMOS technology and seeks applications that require energy-efficient and fast analog-to-digital conversion, with potential for scalability to higher-bit ADCs.

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II. LITERATURE SURVEY

L Koutanali et al. described a low-power Flash ADC architecture that uses the Threshold Inverter Quantization (TIQ) technique. Instead of traditional comparators, they employed cascaded CMOS inverters with transistor sizing to control switching thresholds, removing the need for reference resistor ladders. A logic-based priority encoder generated the digital output based on the highest active comparator. While TIQ was found to be a bit slower than standard Flash ADCs, the study concluded it provided significant advantages in power efficiency and reduced area. They implemented the design with CMOS 180nm technology at a supply voltage of 1.8V, achieving a power consumption of 26.39 µW and a delay of 32.96 µs. [1] E.Chandrasekhar,K.Ramanjaneyulu presented a modified TIQ-based Flash ADC that further improved power efficiency. They pointed out that the excessive use of comparators in Flash ADCs resulted in high power consumption. This issue was tackled by replacing conventional differential comparators and resistive networks with a threshold-compensated TIQ module. Additionally, instead of a standard logic gate-based encoder, they used a MUX-based encoder with transmission gate logic. The design was realized using Tanner EDA tools in 180nm technology. [2] A. Saxena et al. examined how to optimize the comparator design in a 4-bit Flash ADC. The power consumption resulting from a large number of comparators (15 for 4-bit resolution) was a major concern. The study involved designing and comparing different types of comparators to find the most power-efficient option. Among these, a Modified StrongARM Latch Dynamic Comparator yielded the best results, reducing power consumption by 40.6% compared to the Conventional Dynamic Comparator and by 18.68% compared to the standard StrongARM Latch. All designs were simulated using Cadence Virtuoso in both 180nm and 45nm CMOS technologies. [3] Nirali Hemanth Patel analysed the performance of 3-bit and 4-bit Flash ADCs was evaluated, focusing on comparator selection and encoder design. The circuits were designed and initially simulated using the Cadence Spectre simulator under 90nm CMOS technology. The comparator in the design consumed 87 µW, while the total power dissipation for the entire ADCs was 1.05 mW for the 3-bit ADC and 1.984 mW for the 4-bit ADC. The ADC operated at a 25 MHz input frequency with a supply voltage of 1.5V, achieving conversion times of 2.162 ns for the 3-bit and 6.182 ns for the 4-bit configurations. [4] *S.S.Kamate et.al* aimed to create a power-efficient 4-bit Flash ADC designed for biomedical applications where energy considerations are critical. Unlike traditional designs that depend on resistor ladders for reference generation, this study introduced a Sample and Hold (S/H) circuit instead of the R-2R ladder network, leading to significant power savings. They also replaced conventional comparators with simpler, low-power designs and used a priority encoder rather than a traditional thermometer-to-binary encoder. The complete design was implemented using 0.18 µm CMOS technology in Cadence Virtuoso, targeting an analog input range of 1.8V and a signal frequency of 33.20 MHz. [5]

III.METHODOLOGY

A. TIQ Comparator Design

Each TIQ comparator will use a CMOS inverter, with the switching threshold voltage set by the strengths of the NMOS and PMOS transistors. By adjusting the transistor Width-to-Length (W/L) ratios, the switching threshold will be tuned precisely. When we increase the width of the NMOS ,nmos get stronger which shifts the balance between NMOS and PMOS.As a result the inverter switches earlier,thus lowering the switching threshold of the TIQ comparator.When PMOS become stronger the switching threshold shifts toward VDD.This happens beacause now it takes a higher input voltage to make the NMOS overpower the stronger PMOS and flip the output.The target comparator thresholds will be chosen based on the desired quantization levels. For this design, the target switching voltages (Vth) will be: 0.2V, 0.3V, 0.4V, 0.5V, 0.6V, 0.7V, and 0.8V. For each comparator, a CMOS inverter schematic will be created in Cadence Virtuoso. Transistor widths will be adjusted according to the sizing to achieve the intended threshold. For a 3-bit Flash ADC, 7 comparators are used.

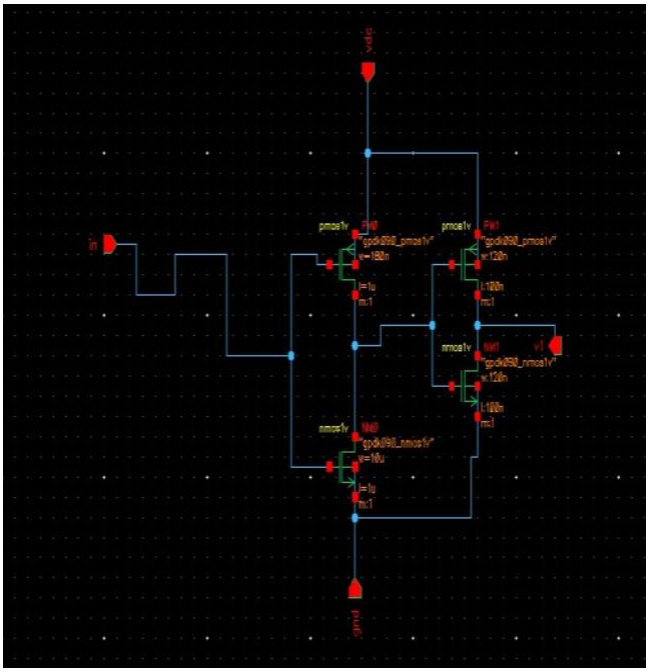

Fig.1.TIQ comparator Design

A total of seven TIQ comparator circuits were designed,each configured with distinct switching threshold voltages.These comparators were simultaneously fed with the input analog signal.The resulting outputs were then passed to a priority encoder,which converted the thermometer coded signals into a corresponding 3-bit digitaloutput,effectively completing the flash ADC operation.This design eliminates the need for a resistor ladder network,reducing power consumption and area.The TIQ-based architecture also alows for high-speed operation due to the fast switching nature of CMOS invertetr.

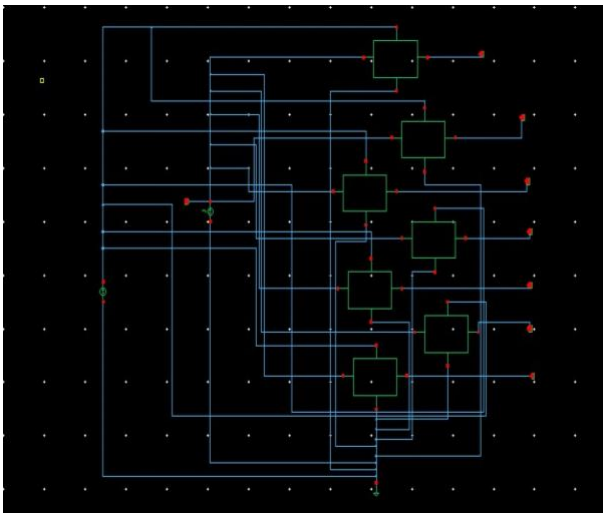

Fig.2. TIQ comparator block

B. Priority Encoder

The priority encoder converts the comparator code into a 3-bit binary output. It prioritizes the highest active comparator output. This logic is implemented using basic AND, OR, and NOT gates, ensuring one unique binary output for any valid input. The final stage provides a 3-bit binary code that corresponds to the voltage range of the analog input.

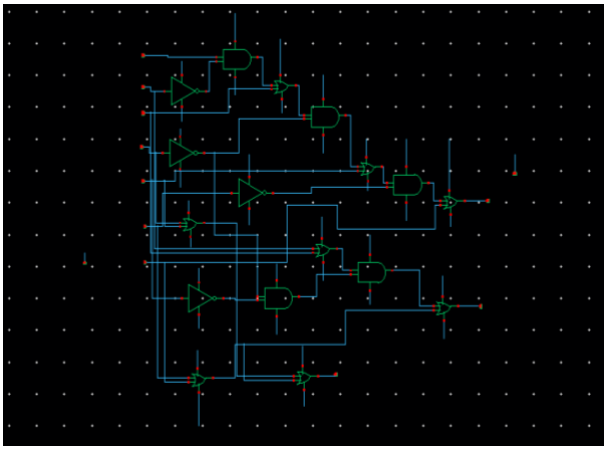

Fig.3.Priority encodet

C.Implementation

Cadence Virtuoso Schematic Editor is used to design the circuit schematic of the Flash ADC, including TIQ comparators and the priority encoder. The processes below show how an analog input signal is changed into a digital output using a sequence of comparators and a priority encoder. This process usually happens in an ADC that uses TIQ comparators. Using comparators and a priority encoder is a straightforward way to change an analog signal into a digital format. TIQ comparators are commonly found in ADCs. First, the analog input goes through several comparators, each set to activate at a different voltage threshold. When conditions are met, each comparator produces a high signal. They all check if the input exceeds their individual thresholds at the same time.

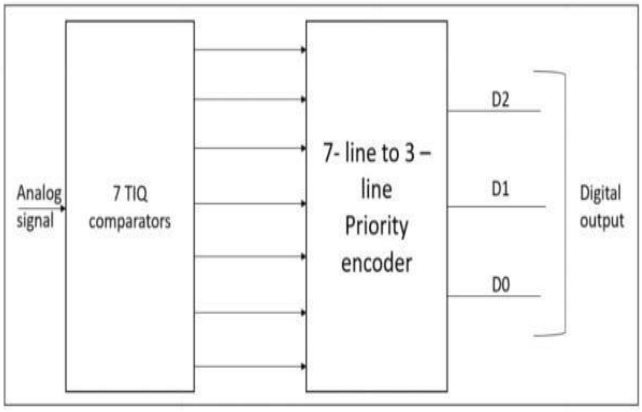

Fig.4. Block diagram of ADC

Next, all the comparator outputs go into a priority encoder, which selects the output from the comparator with the highest threshold voltage that was exceeded. This selection is important for cases when multiple comparators are triggered. It represents the closest estimate of the input voltage's actual magnitude. Then, the priority encoder turns this selection into a digital code, providing a digital representation of the analog input. This clear and effective conversion process completes the cycle, maintaining the details of the analog signal in a digital format that can be used for further electronic processing or analysis.

Table1. Truth table for 3-bit ADC

Analog input	Comparator output							Adc output		
	C7	C6	C5	C4	C3	C2	C1	Y2	Y1	Y0
0-0.2	0	0	0	0	0	0	0	0	0	0
0.2	0	0	0	0	0	0	1	0	0	1
0.3	0	0	0	0	0	1	1	0	1	0
0.4	0	0	0	0	1	1	1	0	1	1
0.5	0	0	0	1	1	1	1	1	0	0
0.6	0	0	1	1	1	1	1	1	0	1
0.7	0	1	1	1	1	1	1	1	1	0
0.8-1	1	1	1	1	1	1	1	1	1	1

IV.RESULTS AND DISCUSSION

The belo waveform shows how the seven TIQ (Threshold Inverter Quantization) comparators behave in the 3-bit Flash ADC. The waveform at the top represents the input analog signal Vin, which is a 1V, 1kHz sinusoidal waveform . Below it, we can see the outputs of the seven comparators (/C1 to /C7). Each comparator has a different switching threshold, ranging from 0.2V to 0.8V. As the input signal rises and crosses these thresholds, the comparator outputs change from logic low to logic high, creating a thermometer code. For instance, when the input voltage is below 0.2V, all comparator outputs stay low, and when it goes above 0.8V, all outputs go high. This behavior shows that the TIQ

comparators effectively detect the analog levels and convert them into the expected thermometer-coded digital form.

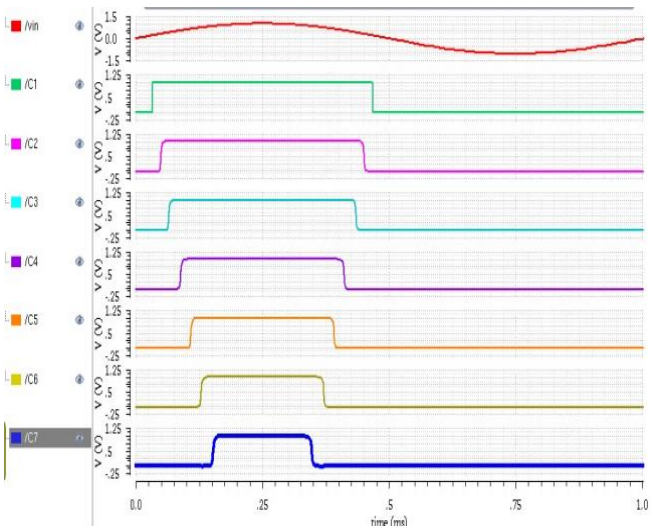

Fig.5.TIQ comparator block result

The below figure shows the output of the priority encoder, which changes the thermometer code from the comparators into a binary code. The encoder inputs are v1 to v7, which match the seven comparator outputs. The encoder looks at these inputs and produces a 3-bit binary output shown as y0, y1, and y2. It picks the highest active (logic high) input and gives its corresponding binary output. For example, if the thermometer input is 1110000, the binary output becomes 100, representing decimal 4. This output changes in time with the thermometer code as the analog input moves through its voltage range. The waveform confirms that the priority encoder operates correctly and can encode the highest comparator index to binary without any glitches or delays.

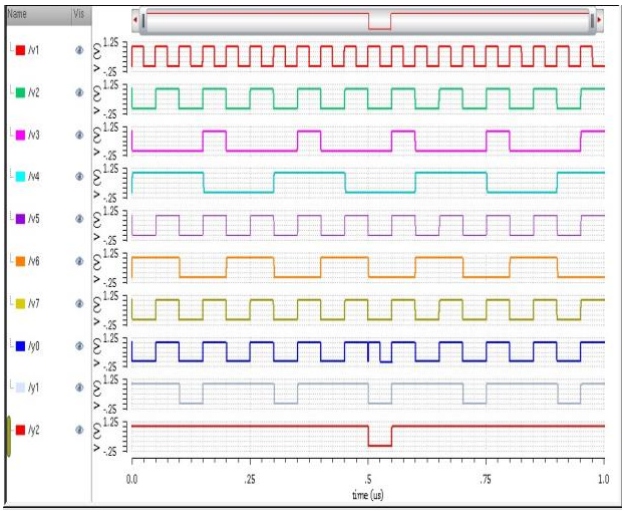

Fig.6. Priority encoder result

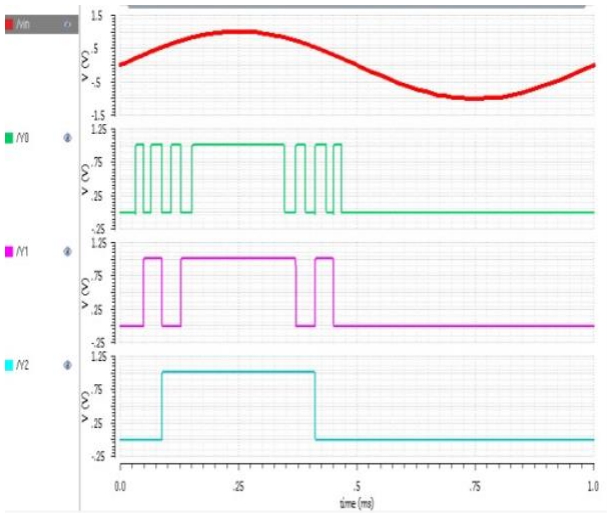

Fig.7. 3-bit ADC result

The final graph illustrates the complete ADC operation, from analog input to final digital output. The analog input Vin is shown again at the top, keeping its sinusoidal shape. The outputs /Y0, /Y1, and /Y2 represent the final 3-bit digital output of the ADC, with /Y0 as the least significant bit (LSB) and /Y2 as the most significant bit (MSB). As the input voltage changes, the digital output switches between the 8 possible binary combinations from 000 to 111. These transitions appear step-like, as expected in a quantized system, and relate to the 8 distinct levels of a 3-bit resolution. The timing and logical accuracy of these transitions confirm that the ADC accurately maps the analog input into its binary digital form. The waveform clearly shows that the ADC responds quickly and precisely to changes in the analog input, proving the successful design and implementation of a high-speed 3-bit Flash ADC using TIQ comparators.

V. CONCLUSION

The proposed 3-bit Flash ADC design using TIQ comparators and a logic gate-based priority encoder provides a strong base for low-power, high-speed analog-to-digital conversion. In the future, this work can expand to higher-bit resolution ADCs by increasing the number of comparators and optimizing the encoder design for speed and area efficiency. Furthermore, new comparator techniques and adaptive W/L tuning through digital calibration can be explored to enhance threshold accuracy and noise resistance. Integration with modern CMOS technologies, such as 65 nm or lower, could further lower power consumption and chip area, making the design appropriate for ultra-low-power IoT and biomedical applications. In addition, adding digital error correction or redundancy techniques could improve performance in challenging or noisy conditions. In this work, we successfully designed and implemented a 3-bit Flash Analog-to-Digital Converter (ADC) using TIQ comparators and a logic gate-based priority encoder. We effectively set the switching thresholds of the comparators by changing the W/L ratios of MOSFETs, ensuring accurate voltage level detection. The complete system was tested with a 1 V, 1 kHz sine wave input and received the

required digital output with a power consumption of about 300 µW and a delay of 40 µs. The design shows the benefits of using TIQ-based architecture in terms of simplicity, speed, and low power, making it a great fit for integration into portable and low-power mixed-signal systems. This work provides a solid foundation for developing higher-resolution ADCs.

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